

London Underground 'What-If' Simulator - Master Backlog

ID	Epic	Item	Description	Sprint	Assignee	Status
1	Project Management, QA & Docs	Set up GitHub repository & branching strategy	Created Git repo; agreed main + feature branches and conventional commits; defined PR + peer-review process	Sprint 1	Jaden	Done
2	Project Management, QA & Docs	Initialise project scaffold (React + Node/Express)	Stood up basic React frontend and Node/Express backend project structure	Sprint 1	Jaden	Done
3	User Interface & UX	Research UI & mapping frameworks	Researched similar tools and UI inspiration; evaluated Leaflet.js vs D3.js for the map	Sprint 1	Sattyaj	Done
4	Project Management, QA & Docs	Set up Agile project board	Created the project board (Google Sheets) and ran the first sprint planning session	Sprint 1	Sinead	Done
5	Project Management, QA & Docs	Project kick-off meeting with supervisor	Scope discussed, requirements clarified, weekly Tuesday meetings confirmed	Sprint 1	All	Done
6	Project Management, QA & Docs	Upload Requirements & User Stories documents	Uploaded requirements and user stories docs; kept project docs under version control	Sprint 1	Jaden	Done
7	Project Management, QA & Docs	Create initial Test Documents	Drafted the initial TestDocuments for the project	Sprint 1	Sattyaj	Done
8	Data Integration & Network Graph	Initial data acquisition (scrape station data)	Wrote scraping/acquisition scripts to gather TfL station data	Sprint 1	Jaden	Done
9	Data Integration & Network Graph	First Neo4j TfL graph iteration	First working iteration building a Neo4j graph of the TfL network	Sprint 1	Jaden	Done
10	Data Integration & Network Graph	Implement TfL API integration	Set up API client against the TfL Unified API; registered API key; scheduled fetch every 5 mins	Sprint 2	Jaden	Done
11	Data Integration & Network Graph	Parse and store fetched data	Transformed raw API data into clean station and line objects; added manual overrides for inconsistent stations	Sprint 2	Jaden	Done
12	Data Integration & Network Graph	Add station uniqueness constraint (Neo4j)	Added constraint-creation query so every station node is unique	Sprint 2	Jaden	Done
13	Data Integration & Network Graph	Error handling & code-quality pass	Added try/catch/finally, comments and general robustness to data-fetch code	Sprint 2	Jaden	Done
14	Project Management, QA & Docs	Write unit tests (data import)	Jest tests for the data-fetch module using mocked API responses; covers parsing and error cases	Sprint 2	Sattyaj	Done
15	Data Integration & Network Graph	Code review (API integration)	Peer review of data integration; minor style fixes applied	Sprint 2	All	Done
16	Data Integration & Network Graph	Map parent-child station relationships	Mapped station components to hubs; validated with known stations (e.g. King's Cross St. Pancras)	Sprint 2	Jaden	Done

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ID	Epic	Item	Description	Sprint	Assignee	Status
17	Data Integration & Network Graph	Build network graph model & populate	Chose adjacency-list graph; populated from TfL data; added distances to edges	Sprint 2	Jaden	Done
18	User Interface & UX	Choose mapping framework (Leaflet.js)	Selected Leaflet.js over D3.js for built-in pan/zoom; first Leaflet map iteration	Sprint 2	Jaden	Done
19	User Interface & UX	Prototype web UI	First prototype UI for the web page with iterative improvements	Sprint 2	Sattyaj	Done
20	Project Management, QA & Docs	Develop Master Test Plan & mock data	Drafted Master Test Plan (unit/integration/UAT, pass-fail criteria); created mock data	Sprint 2	Sattyaj	Done
21	Project Management, QA & Docs	User stories & acceptance criteria document	Wrote user stories and acceptance criteria documentation	Sprint 2	Saf	Done
22	Data Integration & Network Graph	Integrate graph with station hierarchy	Parent station nodes used as routing nodes while preserving platform detail	Sprint 3	Jaden	Done
23	Data Integration & Network Graph	Populate & verify graph connectivity	Imported all lines/connections; Dijkstra script confirmed all nodes reachable; filtered duplicate stations	Sprint 3	Jaden	Done
24	Data Integration & Network Graph	Code review (station hierarchy)	Peer review with renaming/refactoring for clarity	Sprint 3	All	Done
25	User Interface & UX	Render stations on map from Neo4j	Added API request to fetch stations from Neo4j and render them on the Leaflet map	Sprint 3	Jaden	Done
26	User Interface & UX	Show stations & connections on map	First iteration of stations and connections drawn on the map	Sprint 3	Jaden	Done
27	User Interface & UX	Station hover tooltips	Hover tooltips showing station names	Sprint 3	Jaden	Done
28	User Interface & UX	Clickable stations	Clicking selects and highlights a station (to be hooked to routing later)	Sprint 3	Jaden	Done
29	Project Management, QA & Docs	SRS & API documents	Wrote the SRS and API documentation	Sprint 3	Saf	Done
30	Project Management, QA & Docs	TfL Database document	Documented the Neo4j/TfL database design	Sprint 3	Jaden	Done
31	Data Integration & Network Graph	Unit tests (graph construction)	Verified node/edge counts against a small fixed network	Sprint 4	Sattyaj	Done
32	Data Integration & Network Graph	Code review (graph model)	Reviewed graph model; confirmed clean, modular, extensible code	Sprint 4	All	Done
33	User Interface & UX	Draw lines & station connections on map	Rendered station markers and route segments after data load	Sprint 4	Jaden	Done

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ID	Epic	Item	Description	Sprint	Assignee	Status
34	User Interface & UX	Apply TfL line styling	Distinct TfL colours per route and clearer visual separation between lines	Sprint 4	Sinead	Done
35	User Interface & UX	Use Leaflet built-in zoom & pan	Switched zooming/panning to Leaflet's built-in handling	Sprint 4	Jaden	Done
36	Simulation & What-If Scenarios	Implement station/line closure logic	Closing a station/line removes or deactivates its nodes and edges from the routing graph	Sprint 4	Jaden	Done
37	Simulation & What-If Scenarios	UI sync via observer pattern (ID-52)	Observer notification pattern so the UI stays in sync when closures change	Sprint 4	Jaden	Done
38	Project Management, QA & Docs	Update Legal Considerations document	Updated to reflect scope, data usage and TfL usage assumptions	Sprint 4	Saf	Done
39	Project Management, QA & Docs	TfL API endpoint & integration docs	Created TfL API endpoint doc and API integration system doc	Sprint 4	Saf	Done
40	Pathfinding & Routing Engine	Initial pathfinding setup	Initial scaffolding of the pathfinding algorithm	Sprint 4	Jaden	Done
41	Project Management, QA & Docs	Repository maintenance & reorg	Cleaned up documents folder and reorganised repo files	Sprint 4	Jaden	Done
42	Pathfinding & Routing Engine	Implement Dijkstra pathfinding	Dijkstra's algorithm over the network graph; brought onto the frontend	Sprint 5	Jaden	Done
43	Pathfinding & Routing Engine	Unit test pathfinding (sample queries)	Verified correct shortest paths on known routes (e.g. King's Cross to Waterloo)	Sprint 5	Jaden	Done
44	Pathfinding & Routing Engine	Calculate travel times (distance-based)	Route costs computed; distance used for now pending travel-time data	Sprint 5	Jaden	Done
45	Simulation & What-If Scenarios	Unit tests (scenario logic)	Tests for closing a station and finding an alternative or 'no route found'	Sprint 5	Sattyaj	Done
46	User Interface & UX	Station selection & search UI	Station selection interface plus station search, with bug fixes	Sprint 5	Saf	Done
47	Simulation & What-If Scenarios	Toggle station state interactively	Deactivate/activate stations interactively as part of scenario setup	Sprint 5	Sattyaj	Done
48	Simulation & What-If Scenarios	Reset view functionality	Reset clears closure state and restores the full graph/view	Sprint 5	Saf	Done
49	User Interface & UX	Major UI updates	Substantial improvements and refinements to the user interface	Sprint 5	Sattyaj	Done
50	Pathfinding & Routing Engine	Source & parse TfL travel-time data	Sourced TfL travel-time XML and built a CSV of station-to-station times	Sprint 6	Jaden	Done
51	Pathfinding & Routing Engine	Add travel-time logic to Dijkstra	Route costs now based on journey time rather than distance	Sprint 6	Jaden	Done

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ID	Epic	Item	Description	Sprint	Assignee	Status
52	Pathfinding & Routing Engine	Input validation & edge cases	Handles same start/end, unknown station codes and no-route cases with clear messages	Sprint 6	Jaden	Done
53	User Interface & UX	Request route calculation (/api/route)	Frontend calls backend /api/route when stations are selected	Sprint 6	Jaden	Done
54	User Interface & UX	Display route & travel time	Route highlighted in green over the network; travel time shown in a panel	Sprint 6	Jaden	Done
55	User Interface & UX	Clear & update route in UI	Route display clears/refreshes when a new journey or condition changes	Sprint 6	Jaden	Done
56	Simulation & What-If Scenarios	Close entire lines	Added ability to close whole lines	Sprint 6	Sinead	Done
57	User Interface & UX	Line-closure toggle in sidebar	TfL API integration into sidebar with easy line open/close toggles	Sprint 6	Sinead	Done
58	User Interface & UX	Improve route visuals & click mechanics	Made pathfinding more visually interesting; fixed click/double-click mechanics	Sprint 6	Jaden	Done
59	Pathfinding & Routing Engine	Code review (pathfinding)	Reviewed algorithm; suggestion to extract heuristic logic into its own function	Sprint 7	All	Done
60	Simulation & What-If Scenarios	Live partial line closures & disruptions	Map live partial disruptions to specific edges/stations	Sprint 7	Sinead	Done
61	Pathfinding & Routing Engine	Support partial line closures in pathfinding	Pathfinding accounts for partial line closures	Sprint 7	Sinead	Done
62	Simulation & What-If Scenarios	Real-time re-routing on disruptions	Live TfL disruptions trigger closure logic and automatic re-calculation	Sprint 7	Sinead	Done
63	Pathfinding & Routing Engine	Rerouting logic	Reworked rerouting behaviour	Sprint 7	Sattyaj	Done
64	User Interface & UX	Refactor MapCanvas & sidebar	Refactored MapCanvas; sidebar can move off screen	Sprint 7	Sattyaj	Done
65	User Interface & UX	Scenario control panel UI	Side panel listing all lines with open/close toggles, warning colours and a legend	Sprint 7	Sattyaj	Done
66	User Interface & UX	Closed-element styling & route popup	Closed stations show red X; route popup and line names in the box; box moves in what-if mode	Sprint 7	Jaden	Done
67	Simulation & What-If Scenarios	Integrate scenario controls with backend	Toggling a line/station in the UI calls closure logic and updates the map responsively	Sprint 8	Saf	Done
68	Simulation & What-If Scenarios	Station closure fixes	Several fixes to station-closure behaviour (including a revert and full fix)	Sprint 8	Saf	Done
69	Pathfinding & Routing Engine	Pathfinding favours staying on a line	Algorithm prefers staying on the same line over constant switching	Sprint 8	Jaden	Done

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ID	Epic	Item	Description	Sprint	Assignee	Status
70	Live Data & Train Display	Live + fallback train movements and UI	Added live train movements with a fallback, plus supporting UI	Sprint 8	Sinead	Done
71	Simulation & What-If Scenarios	Evaluate impact on routes	Confirmed new route queries use the updated graph after closures	Sprint 8	Saf	Done
72	Project Management, QA & Docs	Unit/integration tests (disruption UI)	Added tests so closing a station removes it from subsequent routes end-to-end	Sprint 8	Sattyaj	Done
73	Project Management, QA & Docs	Update Technical Report (disruption & what-if)	Documented manual and live closures and what-if functionality with screenshots	Sprint 8	All	Done
74	Project Management, QA & Docs	Full integration testing & defect fixes	End-to-end tests (map, station select, route, closures, re-route); fixed minor defects	Sprint 9	Jaden	Done
75	Data Integration & Network Graph	Refactor functions into separate files	Split functions across files and tidied folder structure	Sprint 9	Jaden	Done
76	Live Data & Train Display	Disable live feeds & refactor status	Disabled live feeds where needed and refactored status handling	Sprint 9	Sinead	Done
77	User Interface & UX	Refactor map layer styling	Refactored map layer styling and fixed a types import	Sprint 9	Sinead	Done
78	Live Data & Train Display	Add delay information	Added line delay information	Sprint 9	Sattyaj	Done
79	User Interface & UX	Dynamic X-icon size & clear-on-reclick	X icon resizes dynamically; clicking the start station again clears it	Sprint 9	Jaden	Done
80	Project Management, QA & Docs	Design project poster & abstract	A1 poster with title, abstract (~150 words), architecture diagram and map screenshot	Sprint 9	Sinead	Done
81	Project Management, QA & Docs	Print draft & review poster	Printed A4 draft, reviewed as a team, finalised the poster PDF	Sprint 9	All	Done
82	Project Management, QA & Docs	Regroup after Easter break	Re-familiarised with the codebase/docs and re-agreed priorities for the term	Sprint 10	All	Done
83	Live Data & Train Display	Select which line's trains to display	User can choose which line's trains to show rather than all at once	Sprint 10	Jaden	Done
84	Live Data & Train Display	Refactor train ID helpers	Moved helper functions out of useTrainMovements into trainIdUtils	Sprint 10	Jaden	Done
85	Live Data & Train Display	Stop unnecessary re-renders on feed update	Fixed a child-parent update loop firing on every feed update	Sprint 10	Jaden	Done
86	Live Data & Train Display	Move live train API call to backend	Live-train TfL call moved server-side so the API key isn't exposed	Sprint 11	Jaden	Done
87	Live Data & Train Display	Add trackable-trains counter	Variable to track how many trains are live	Sprint 11	Jaden	Done
88	Live Data & Train Display	Refactor train graph helpers	Moved several functions into trainGraphUtils	Sprint 11	Jaden	Done
89	User Interface & UX	Improve map & line loading	Improved line loading and general map loading performance	Sprint 11	Sattyaj	Done

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ID	Epic	Item	Description	Sprint	Assignee	Status
90	User Interface & UX	Add max map bound	Max bound so users can't pan endlessly away from London	Sprint 11	Sattyaj	Done
91	Simulation & What-If Scenarios	Fix station closure bug	Fixed an outstanding station-closure bug	Sprint 11	Saf	Done
92	User Interface & UX	Extract state out of MapCanvas	Pulled what-if closures, station search, live closures, line delays, HUD and route panel into focused files	Sprint 12	Jaden	Done
93	User Interface & UX	Remove duplicate tile rendering	Removed duplicate tile rendering in LeafletMap	Sprint 12	Jaden	Done
94	Live Data & Train Display	Predict next leg of train journeys	Predict next part of each journey from API data with estimated values and corrected location text	Sprint 12	Jaden	Done
95	Live Data & Train Display	Add Jubilee & Metropolitan trains	Added Jubilee and Metropolitan line trains to the graph	Sprint 12	Jaden	Done
96	Live Data & Train Display	Refactor & document train modules	Split useTrainMovements into files; added comments across train modules	Sprint 12	Jaden	Done
97	User Interface & UX	Improve mobile UI	Improvements to the UI for mobile use	Sprint 12	Sattyaj	Done
98	Project Management, QA & Docs	Add new test documents	Added new test documents and tidied existing ones	Sprint 12	Sattyaj	Done
99	Live Data & Train Display	Train icon change + tests	Updated the train icon and added tests	Sprint 12	Saf	Done
100	Live Data & Train Display	Render trains on canvas & cache icons	Switched train rendering to canvas and cached icons for a big performance gain	Sprint 13	Jaden	Done
101	Live Data & Train Display	Fix memo-breaking callbacks & tune ticks	Fixed callbacks into StationLayer; set animation ticks to 1000ms for smoother movement	Sprint 13	Jaden	Done
102	Live Data & Train Display	Dynamically resize trains with zoom	Trains shrink/enlarge with zoom level; tuned zoom increments	Sprint 13	Jaden	Done
103	Project Management, QA & Docs	Add Docker setup & test scripts	Docker setup plus Mac/PowerShell scripts to run all tests; updated Babel/Jest	Sprint 13	Jaden	Done
104	User Interface & UX	Search clear button & arrow-key nav	Added a clear button and arrow-key navigation to station search	Sprint 13	Saf	Done
105	User Interface & UX	Glow searched station	A searched station glows so it is easy to spot	Sprint 13	Saf	Done
106	User Interface & UX	Improve train status box position	Repositioned the train status box	Sprint 13	Jaden	Done
107	User Interface & UX	Mobile UI & reorganisation	Mobile UI work and reorganisation; reset view now also resets routes	Sprint 13	Sinead	Done
108	User Interface & UX	Add animations & minor UI	Added animations and minor UI changes	Sprint 13	Sattyaj	Done
109	Project Management, QA & Docs	Prepare user study materials	Drafted task sheet, questionnaire and consent sheet for the study	Sprint 13	All	Done

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ID	Epic	Item	Description	Sprint	Assignee	Status
110	Project Management, QA & Docs	Finalise study materials & recruit	Finalised study materials and recruited participants	Sprint 14	All	Done
111	Project Management, QA & Docs	Launch user study	Launched the user study on 14 May with guided tasks and a questionnaire	Sprint 14	All	Done
112	User Interface & UX	Improve search/route interaction	Improved search/route interaction and map controls copy; added route/closure signposting	Sprint 14	Sinead	Done
113	User Interface & UX	Refine search focus & collapsible controls	Refined search focus and added collapsible map controls to reduce clutter	Sprint 14	Sinead	Done
114	User Interface & UX	Render trains beneath stations	Trains render beneath stations so a tap/click hits the station first	Sprint 14	Sinead	Done
115	User Interface & UX	Align search box with route info box	Moved the search box to line up with the route info box	Sprint 14	Jaden	Done
116	Simulation & What-If Scenarios	Begin save/load feature	Started the save/load scenarios feature	Sprint 14	Saf	Done
117	Project Management, QA & Docs	Finish study sessions & collate responses	Completed study sessions and began collating questionnaire/observation data	Sprint 15	All	Done
118	User Interface & UX	Add light/dark theme tokens & toggle	Added light/dark tokens and a slider; applied light mode across box components	Sprint 15	Jaden	Done
119	User Interface & UX	Fix theming hydration errors	Fixed hydration issues caused by the light-mode slider and line-info button	Sprint 15	Jaden	Done
120	Simulation & What-If Scenarios	Complete save/load feature	Users can store and reload hypothetical disruption scenarios	Sprint 15	Saf	Done
121	User Interface & UX	Bring what-if mode onto main screen	Moved what-if mode to the main screen and made it easier to see	Sprint 15	Saf	Done
122	User Interface & UX	Centre map when routing	Map now centres when a route is calculated	Sprint 15	Saf	Done
123	Project Management, QA & Docs	Clean file & folder structure	Untracked DS_Store files, moved scripts to a scripts folder, tidied the app folder	Sprint 15	Jaden	Done
124	Project Management, QA & Docs	Collate user study results	Pulled out priorities: clearer search, more obvious what-if mode, better map readability	Sprint 16	Sinead	Done
125	User Interface & UX	Apply user study fixes	Polished map search and route panel interactions from study feedback	Sprint 16	Sinead	Done
126	User Interface & UX	Refactor map control panel	Split toggles, interaction-mode toggle, saved scenarios, train controls, line status and network summary into files	Sprint 16	Jaden	Done

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ID	Epic	Item	Description	Sprint	Assignee	Status
127	User Interface & UX	Refactor MapSearchBox	Split inputs, results list, focused-station panel and collapsed button into separate files	Sprint 16	Jaden	Done
128	User Interface & UX	Build guided tour of the UI	Working guided tour for first-time users	Sprint 16	Sattyaj	Done
129	Data Integration & Network Graph	Route TfL API via internal Next.js API	All TfL API calls now go only through the internal Next.js API	Sprint 16	Jaden	Done
130	Project Management, QA & Docs	Merge requirements docs & start README	Merged requirements docs, cleaned the API document and started the README	Sprint 16	Jaden	Done
131	Project Management, QA & Docs	Local Neo4j via Docker for tests	Added a local Neo4j Docker DB and made tests runnable via Docker for easier marking	Sprint 16	Jaden	Done
132	Project Management, QA & Docs	Add GitLab CI pipeline	Added a GitLab CI pipeline for backend and frontend tests	Sprint 16	Sinead	Done
133	Project Management, QA & Docs	Expand test coverage	Expanded frontend and routing test coverage	Sprint 16	Sinead	Done
134	Project Management, QA & Docs	Upload study report, consent sheet & poster	Uploaded the user study report, consent sheet and poster/abstract materials	Sprint 16	Sinead	Done
135	Project Management, QA & Docs	Fix lint errors & tidy repo	Fixed frontend lint errors and tidied the repo for hand-in	Sprint 16	Jaden	Done
136	Project Management, QA & Docs	Documentation updates	Document updates and naming-convention fixes	Sprint 16	Saf	Done

Sprint 1 (Jan 12 - Jan 16, 2026)

Goal: Project kick-off, set up infrastructure, begin TfL API integration

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	1	Set up GitHub repository & branching strategy	Created Git repo; agreed main + feature branches and conventional commits; defined PR + peer-review process	Project Management, QA & Docs	Jaden	Done
TRUE	2	Initialise project scaffold (React + Node/Express)	Stood up basic React frontend and Node/Express backend project structure	Project Management, QA & Docs	Jaden	Done
TRUE	3	Research UI & mapping frameworks	Researched similar tools and UI inspiration; evaluated Leaflet.js vs D3.js for the map	User Interface & UX	Sattyaj	Done
TRUE	4	Set up Agile project board	Created the project board (Google Sheets) and ran the first sprint planning session	Project Management, QA & Docs	Sinead	Done
TRUE	5	Project kick-off meeting with supervisor	Scope discussed, requirements clarified, weekly Tuesday meetings confirmed	Project Management, QA & Docs	All	Done
TRUE	6	Upload Requirements & User Stories documents	Uploaded requirements and user stories docs; kept project docs under version control	Project Management, QA & Docs	Jaden	Done
TRUE	7	Create initial Test Documents	Drafted the initial TestDocuments for the project	Project Management, QA & Docs	Sattyaj	Done
TRUE	8	Initial data acquisition (scrape station data)	Wrote scraping/acquisition scripts to gather TfL station data	Data Integration & Network Graph	Jaden	Done
TRUE	9	First Neo4j TfL graph iteration	First working iteration building a Neo4j graph of the TfL network	Data Integration & Network Graph	Jaden	Done

Sprint 2 (Jan 19 - Jan 23, 2026)

Goal: TfL API integration, data import unit tests, station hub model, initial map rendering, finalise plan

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	10	Implement TfL API integration	Set up API client against the TfL Unified API; registered API key; scheduled fetch every 5 mins	Data Integration & Network Graph	Jaden	Done
TRUE	11	Parse and store fetched data	Transformed raw API data into clean station and line objects; added manual overrides for inconsistent stations	Data Integration & Network Graph	Jaden	Done
TRUE	12	Add station uniqueness constraint (Neo4j)	Added constraint-creation query so every station node is unique	Data Integration & Network Graph	Jaden	Done
TRUE	13	Error handling & code-quality pass	Added try/catch/finally, comments and general robustness to data-fetch code	Data Integration & Network Graph	Jaden	Done
TRUE	14	Write unit tests (data import)	Jest tests for the data-fetch module using mocked API responses; covers parsing and error cases	Project Management, QA & Docs	Sattyaj	Done
TRUE	15	Code review (API integration)	Peer review of data integration; minor style fixes applied	Data Integration & Network Graph	All	Done
TRUE	16	Map parent-child station relationships	Mapped station components to hubs; validated with known stations (e.g. King's Cross St. Pancras)	Data Integration & Network Graph	Jaden	Done
TRUE	17	Build network graph model & populate	Chose adjacency-list graph; populated from TfL data; added distances to edges	Data Integration & Network Graph	Jaden	Done
TRUE	18	Choose mapping framework (Leaflet.js)	Selected Leaflet.js over D3.js for built-in pan/zoom; first Leaflet map iteration	User Interface & UX	Jaden	Done
TRUE	19	Prototype web UI	First prototype UI for the web page with iterative improvements	User Interface & UX	Sattyaj	Done
TRUE	20	Develop Master Test Plan & mock data	Drafted Master Test Plan (unit/integration/UAT, pass-fail criteria); created mock data	Project Management, QA & Docs	Sattyaj	Done
TRUE	21	User stories & acceptance criteria document	Wrote user stories and acceptance criteria documentation	Project Management, QA & Docs	Saf	Done

Sprint 3 (Jan 26 - Jan 30, 2026)

Goal: Graph construction, integrate station hierarchy, begin interactive map features

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	22	Integrate graph with station hierarchy	Parent station nodes used as routing nodes while preserving platform detail	Data Integration & Network Graph	Jaden	Done
TRUE	23	Populate & verify graph connectivity	Imported all lines/connections; Dijkstra script confirmed all nodes reachable; filtered duplicate stations	Data Integration & Network Graph	Jaden	Done
TRUE	24	Code review (station hierarchy)	Peer review with renaming/refactoring for clarity	Data Integration & Network Graph	All	Done
TRUE	25	Render stations on map from Neo4j	Added API request to fetch stations from Neo4j and render them on the Leaflet map	User Interface & UX	Jaden	Done
TRUE	26	Show stations & connections on map	First iteration of stations and connections drawn on the map	User Interface & UX	Jaden	Done
TRUE	27	Station hover tooltips	Hover tooltips showing station names	User Interface & UX	Jaden	Done
TRUE	28	Clickable stations	Clicking selects and highlights a station (to be hooked to routing later)	User Interface & UX	Jaden	Done
TRUE	29	SRS & API documents	Wrote the SRS and API documentation	Project Management, QA & Docs	Saf	Done
TRUE	30	TfL Database document	Documented the Neo4j/TfL database design	Project Management, QA & Docs	Jaden	Done

Sprint 4 (Feb 2 - Feb 6, 2026)

Goal: Render lines/connections, closure simulation logic, UI testing & refactor, pathfinding setup

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	31	Unit tests (graph construction)	Verified node/edge counts against a small fixed network	Data Integration & Network Graph	Sattyaj	Done
TRUE	32	Code review (graph model)	Reviewed graph model; confirmed clean, modular, extensible code	Data Integration & Network Graph	All	Done
TRUE	33	Draw lines & station connections on map	Rendered station markers and route segments after data load	User Interface & UX	Jaden	Done
TRUE	34	Apply TfL line styling	Distinct TfL colours per route and clearer visual separation between lines	User Interface & UX	Sinead	Done
TRUE	35	Use Leaflet built-in zoom & pan	Switched zooming/panning to Leaflet's built-in handling	User Interface & UX	Jaden	Done
TRUE	36	Implement station/line closure logic	Closing a station/line removes or deactivates its nodes and edges from the routing graph	Simulation & What-If Scenarios	Jaden	Done
TRUE	37	UI sync via observer pattern (ID-52)	Observer notification pattern so the UI stays in sync when closures change	Simulation & What-If Scenarios	Jaden	Done
TRUE	38	Update Legal Considerations document	Updated to reflect scope, data usage and TfL usage assumptions	Project Management, QA & Docs	Saf	Done
TRUE	39	TfL API endpoint & integration docs	Created TfL API endpoint doc and API integration system doc	Project Management, QA & Docs	Saf	Done
TRUE	40	Initial pathfinding setup	Initial scaffolding of the pathfinding algorithm	Pathfinding & Routing Engine	Jaden	Done
TRUE	41	Repository maintenance & reorg	Cleaned up documents folder and reorganised repo files	Project Management, QA & Docs	Jaden	Done

Sprint 5 (Feb 9 - Feb 13, 2026)

Goal: Implement core pathfinding algorithm, travel times, station selection UI

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	42	Implement Dijkstra pathfinding	Dijkstra's algorithm over the network graph; brought onto the frontend	Pathfinding & Routing Engine	Jaden	Done
TRUE	43	Unit test pathfinding (sample queries)	Verified correct shortest paths on known routes (e.g. King's Cross to Waterloo)	Pathfinding & Routing Engine	Jaden	Done
TRUE	44	Calculate travel times (distance-based)	Route costs computed; distance used for now pending travel-time data	Pathfinding & Routing Engine	Jaden	Done
TRUE	45	Unit tests (scenario logic)	Tests for closing a station and finding an alternative or 'no route found'	Simulation & What-If Scenarios	Sattyaj	Done
TRUE	46	Station selection & search UI	Station selection interface plus station search, with bug fixes	User Interface & UX	Saf	Done
TRUE	47	Toggle station state interactively	Deactivate/activate stations interactively as part of scenario setup	Simulation & What-If Scenarios	Sattyaj	Done
TRUE	48	Reset view functionality	Reset clears closure state and restores the full graph/view	Simulation & What-If Scenarios	Saf	Done
TRUE	49	Major UI updates	Substantial improvements and refinements to the user interface	User Interface & UX	Sattyaj	Done

Sprint 6 (Feb 16 - Feb 20, 2026)

Goal: Display route & travel time, dynamic re-routing on closures, integrate TfL travel-time data

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	50	Source & parse TfL travel-time data	Sourced TfL travel-time XML and built a CSV of station-to-station times	Pathfinding & Routing Engine	Jaden	Done
TRUE	51	Add travel-time logic to Dijkstra	Route costs now based on journey time rather than distance	Pathfinding & Routing Engine	Jaden	Done
TRUE	52	Input validation & edge cases	Handles same start/end, unknown station codes and no-route cases with clear messages	Pathfinding & Routing Engine	Jaden	Done
TRUE	53	Request route calculation (/api/route)	Frontend calls backend /api/route when stations are selected	User Interface & UX	Jaden	Done
TRUE	54	Display route & travel time	Route highlighted in green over the network; travel time shown in a panel	User Interface & UX	Jaden	Done
TRUE	55	Clear & update route in UI	Route display clears/refreshes when a new journey or condition changes	User Interface & UX	Jaden	Done
TRUE	56	Close entire lines	Added ability to close whole lines	Simulation & What-If Scenarios	Sinead	Done
TRUE	57	Line-closure toggle in sidebar	TfL API integration into sidebar with easy line open/close toggles	User Interface & UX	Sinead	Done
TRUE	58	Improve route visuals & click mechanics	Made pathfinding more visually interesting; fixed click/double-click mechanics	User Interface & UX	Jaden	Done

Sprint 7 (Feb 23 - Feb 27, 2026)

Goal: Code reviews, live partial closures, scenario control panel UI, visual continuity

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	59	Code review (pathfinding)	Reviewed algorithm; suggestion to extract heuristic logic into its own function	Pathfinding & Routing Engine	All	Done
TRUE	60	Live partial line closures & disruptions	Map live partial disruptions to specific edges/stations	Simulation & What-If Scenarios	Sinead	Done
TRUE	61	Support partial line closures in pathfinding	Pathfinding accounts for partial line closures	Pathfinding & Routing Engine	Sinead	Done
TRUE	62	Real-time re-routing on disruptions	Live TfL disruptions trigger closure logic and automatic re-calculation	Simulation & What-If Scenarios	Sinead	Done
TRUE	63	Rerouting logic	Reworked rerouting behaviour	Pathfinding & Routing Engine	Sattyaj	Done
TRUE	64	Refactor MapCanvas & sidebar	Refactored MapCanvas; sidebar can move off screen	User Interface & UX	Sattyaj	Done
TRUE	65	Scenario control panel UI	Side panel listing all lines with open/close toggles, warning colours and a legend	User Interface & UX	Sattyaj	Done
TRUE	66	Closed-element styling & route popup	Closed stations show red X; route popup and line names in the box; box moves in what-if mode	User Interface & UX	Jaden	Done

Sprint 8 (Mar 2 - Mar 6, 2026)

Goal: Full integration of scenario controls, disruption UI tests, technical report updates

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	67	Integrate scenario controls with backend	Toggling a line/station in the UI calls closure logic and updates the map responsively	Simulation & What-If Scenarios	Saf	Done
TRUE	68	Station closure fixes	Several fixes to station-closure behaviour (including a revert and full fix)	Simulation & What-If Scenarios	Saf	Done
TRUE	69	Pathfinding favours staying on a line	Algorithm prefers staying on the same line over constant switching	Pathfinding & Routing Engine	Jaden	Done
TRUE	70	Live + fallback train movements and UI	Added live train movements with a fallback, plus supporting UI	Live Data & Train Display	Sinead	Done
TRUE	71	Evaluate impact on routes	Confirmed new route queries use the updated graph after closures	Simulation & What-If Scenarios	Saf	Done
TRUE	72	Unit/integration tests (disruption UI)	Added tests so closing a station removes it from subsequent routes end-to-end	Project Management, QA & Docs	Sattyaj	Done
TRUE	73	Update Technical Report (disruption & what-if)	Documented manual and live closures and what-if functionality with screenshots	Project Management, QA & Docs	All	Done

Sprint 9 (Mar 9 - Mar 13, 2026)

Goal: Integration testing, project poster and abstract for the Project Fair

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	74	Full integration testing & defect fixes	End-to-end tests (map, station select, route, closures, re-route); fixed minor defects	Project Management, QA & Docs	Jaden	Done
TRUE	75	Refactor functions into separate files	Split functions across files and tidied folder structure	Data Integration & Network Graph	Jaden	Done
TRUE	76	Disable live feeds & refactor status	Disabled live feeds where needed and refactored status handling	Live Data & Train Display	Sinead	Done
TRUE	77	Refactor map layer styling	Refactored map layer styling and fixed a types import	User Interface & UX	Sinead	Done
TRUE	78	Add delay information	Added line delay information	Live Data & Train Display	Sattyaj	Done
TRUE	79	Dynamic X-icon size & clear-on-reclick	X icon resizes dynamically; clicking the start station again clears it	User Interface & UX	Jaden	Done
TRUE	80	Design project poster & abstract	A1 poster with title, abstract (~150 words), architecture diagram and map screenshot	Project Management, QA & Docs	Sinead	Done
TRUE	81	Print draft & review poster	Printed A4 draft, reviewed as a team, finalised the poster PDF	Project Management, QA & Docs	All	Done

Sprint 10 (Apr 13 - Apr 17, 2026)

Goal: First week back after Easter: regroup, re-familiarise, resume train display work

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	82	Regroup after Easter break	Re-familiarised with the codebase/docs and re-agreed priorities for the term	Project Management, QA & Docs	All	Done
TRUE	83	Select which line's trains to display	User can choose which line's trains to show rather than all at once	Live Data & Train Display	Jaden	Done
TRUE	84	Refactor train ID helpers	Moved helper functions out of useTrainMovements into trainIdUtils	Live Data & Train Display	Jaden	Done
TRUE	85	Stop unnecessary re-renders on feed update	Fixed a child-parent update loop firing on every feed update	Live Data & Train Display	Jaden	Done

Sprint 11 (Apr 20 - Apr 24, 2026)

Goal: Backend train data and map loading improvements

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	86	Move live train API call to backend	Live-train TfL call moved server-side so the API key isn't exposed	Live Data & Train Display	Jaden	Done
TRUE	87	Add trackable-trains counter	Variable to track how many trains are live	Live Data & Train Display	Jaden	Done
TRUE	88	Refactor train graph helpers	Moved several functions into trainGraphUtils	Live Data & Train Display	Jaden	Done
TRUE	89	Improve map & line loading	Improved line loading and general map loading performance	User Interface & UX	Sattyaj	Done
TRUE	90	Add max map bound	Max bound so users can't pan endlessly away from London	User Interface & UX	Sattyaj	Done
TRUE	91	Fix station closure bug	Fixed an outstanding station-closure bug	Simulation & What-If Scenarios	Saf	Done

Sprint 12 (Apr 27 - May 1, 2026)

Goal: Major MapCanvas refactor, smarter train movement prediction, mobile UI

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	92	Extract state out of MapCanvas	Pulled what-if closures, station search, live closures, line delays, HUD and route panel into focused files	User Interface & UX	Jaden	Done
TRUE	93	Remove duplicate tile rendering	Removed duplicate tile rendering in LeafletMap	User Interface & UX	Jaden	Done
TRUE	94	Predict next leg of train journeys	Predict next part of each journey from API data with estimated values and corrected location text	Live Data & Train Display	Jaden	Done
TRUE	95	Add Jubilee & Metropolitan trains	Added Jubilee and Metropolitan line trains to the graph	Live Data & Train Display	Jaden	Done
TRUE	96	Refactor & document train modules	Split useTrainMovements into files; added comments across train modules	Live Data & Train Display	Jaden	Done
TRUE	97	Improve mobile UI	Improvements to the UI for mobile use	User Interface & UX	Sattyaj	Done
TRUE	98	Add new test documents	Added new test documents and tidied existing ones	Project Management, QA & Docs	Sattyaj	Done
TRUE	99	Train icon change + tests	Updated the train icon and added tests	Live Data & Train Display	Saf	Done

Sprint 13 (May 4 - May 8, 2026)

Goal: Train rendering performance pass, search/interaction polish, prepare user study

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	100	Render trains on canvas & cache icons	Switched train rendering to canvas and cached icons for a big performance gain	Live Data & Train Display	Jaden	Done
TRUE	101	Fix memo-breaking callbacks & tune ticks	Fixed callbacks into StationLayer; set animation ticks to 1000ms for smoother movement	Live Data & Train Display	Jaden	Done
TRUE	102	Dynamically resize trains with zoom	Trains shrink/enlarge with zoom level; tuned zoom increments	Live Data & Train Display	Jaden	Done
TRUE	103	Add Docker setup & test scripts	Docker setup plus Mac/PowerShell scripts to run all tests; updated Babel/Jest	Project Management, QA & Docs	Jaden	Done
TRUE	104	Search clear button & arrow-key nav	Added a clear button and arrow-key navigation to station search	User Interface & UX	Saf	Done
TRUE	105	Glow searched station	A searched station glows so it is easy to spot	User Interface & UX	Saf	Done
TRUE	106	Improve train status box position	Repositioned the train status box	User Interface & UX	Jaden	Done
TRUE	107	Mobile UI & reorganisation	Mobile UI work and reorganisation; reset view now also resets routes	User Interface & UX	Sinead	Done
TRUE	108	Add animations & minor UI	Added animations and minor UI changes	User Interface & UX	Sattyaj	Done
TRUE	109	Prepare user study materials	Drafted task sheet, questionnaire and consent sheet for the study	Project Management, QA & Docs	All	Done

Sprint 14 (May 11 - May 15, 2026)

Goal: Launch user study, search/route polish, begin save/load feature

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	110	Finalise study materials & recruit	Finalised study materials and recruited participants	Project Management, QA & Docs	All	Done
TRUE	111	Launch user study	Launched the user study on 14 May with guided tasks and a questionnaire	Project Management, QA & Docs	All	Done
TRUE	112	Improve search/route interaction	Improved search/route interaction and map controls copy; added route/closure signposting	User Interface & UX	Sinead	Done
TRUE	113	Refine search focus & collapsible controls	Refined search focus and added collapsible map controls to reduce clutter	User Interface & UX	Sinead	Done
TRUE	114	Render trains beneath stations	Trains render beneath stations so a tap/click hits the station first	User Interface & UX	Sinead	Done
TRUE	115	Align search box with route info box	Moved the search box to line up with the route info box	User Interface & UX	Jaden	Done
TRUE	116	Begin save/load feature	Started the save/load scenarios feature	Simulation & What-If Scenarios	Saf	Done

Sprint 15 (May 18 - May 22, 2026)

Goal: Finish user study, light/dark mode, complete save/load feature

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	117	Finish study sessions & collate responses	Completed study sessions and began collating questionnaire/observation data	Project Management, QA & Docs	All	Done
TRUE	118	Add light/dark theme tokens & toggle	Added light/dark tokens and a slider; applied light mode across box components	User Interface & UX	Jaden	Done
TRUE	119	Fix theming hydration errors	Fixed hydration issues caused by the light-mode slider and line-info button	User Interface & UX	Jaden	Done
TRUE	120	Complete save/load feature	Users can store and reload hypothetical disruption scenarios	Simulation & What-If Scenarios	Saf	Done
TRUE	121	Bring what-if mode onto main screen	Moved what-if mode to the main screen and made it easier to see	User Interface & UX	Saf	Done
TRUE	122	Centre map when routing	Map now centres when a route is calculated	User Interface & UX	Saf	Done
TRUE	123	Clean file & folder structure	Untracked DS_Store files, moved scripts to a scripts folder, tidied the app folder	Project Management, QA & Docs	Jaden	Done

Sprint 16 (May 25 - May 29, 2026)

Goal: Act on user study findings, codebase clean-up (control panel, guided tour), DevEx (docs, Docker, CI)

Done	ID	Task	Description	Epic	Assignee	Status
TRUE	124	Collate user study results	Pulled out priorities: clearer search, more obvious what-if mode, better map readability	Project Management, QA & Docs	Sinead	Done
TRUE	125	Apply user study fixes	Polished map search and route panel interactions from study feedback	User Interface & UX	Sinead	Done
TRUE	126	Refactor map control panel	Split toggles, interaction-mode toggle, saved scenarios, train controls, line status and network summary into files	User Interface & UX	Jaden	Done
TRUE	127	Refactor MapSearchBox	Split inputs, results list, focused-station panel and collapsed button into separate files	User Interface & UX	Jaden	Done
TRUE	128	Build guided tour of the UI	Working guided tour for first-time users	User Interface & UX	Sattyaj	Done
TRUE	129	Route TfL API via internal Next.js API	All TfL API calls now go only through the internal Next.js API	Data Integration & Network Graph	Jaden	Done
TRUE	130	Merge requirements docs & start README	Merged requirements docs, cleaned the API document and started the README	Project Management, QA & Docs	Jaden	Done
TRUE	131	Local Neo4j via Docker for tests	Added a local Neo4j Docker DB and made tests runnable via Docker for easier marking	Project Management, QA & Docs	Jaden	Done
TRUE	132	Add GitLab CI pipeline	Added a GitLab CI pipeline for backend and frontend tests	Project Management, QA & Docs	Sinead	Done
TRUE	133	Expand test coverage	Expanded frontend and routing test coverage	Project Management, QA & Docs	Sinead	Done
TRUE	134	Upload study report, consent sheet & poster	Uploaded the user study report, consent sheet and poster/abstract materials	Project Management, QA & Docs	Sinead	Done
TRUE	135	Fix lint errors & tidy repo	Fixed frontend lint errors and tidied the repo for hand-in	Project Management, QA & Docs	Jaden	Done
TRUE	136	Documentation updates	Document updates and naming-convention fixes	Project Management, QA & Docs	Saf	Done

Individual Contributions Summary

Primary-owner counts per sprint, from delivered backlog items ('All' = whole-team items).

Member	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12	S13	S14	S15	S16	Total
Jaden	5	7	7	6	3	7	1	1	3	3	3	5	5	1	3	6	66
Sattyaj	2	3		1	3		3	1	1		2	2	1			1	20
Sinead	1			1		2	3	1	3				1	3		5	20
Saf		1	1	2	2			3			1	1	2	1	3	1	18
All	1	1	1	1			1	1	1	1			1	2	1		12
Total	9	12	9	11	8	9	8	7	8	4	6	8	10	7	7	13	136